

Technical Explanation for Beginners

Understanding the Tools, Algorithms, and
Workflow Used in the ADHD Behavioral
Prediction AI Project

By Christos Emmanouil Loukas 2026



Python

What is it?

Python is a high-level programming language widely used in artificial intelligence, machine learning, and scientific computing.

Why was it used?

Python has one of the richest ecosystems for data science, allowing the entire workflow—from loading data to training models and generating explanations—to be implemented in a single language.

Pandas

What is it?

Pandas is a library for manipulating structured data.

Think of it as an advanced spreadsheet inside Python.

It allows you to:

- load CSV/TSV files
- inspect datasets
- filter rows
- clean missing values
- create new variables
- save processed datasets

How did we use it?

We used Pandas to:

- Load the ADHD-200 dataset
- Explore the variables
- Remove unwanted diagnosis categories
- Create the binary ADHD variable
- Save the cleaned dataset

Without Pandas, preprocessing would have been extremely difficult.

NumPy

What is it?

NumPy is Python's numerical computing library.

It stores data as efficient mathematical arrays.

Why is it important?

Machine learning algorithms don't work directly on spreadsheets.

They work on numerical matrices.

NumPy provides:

- fast mathematical operations
- linear algebra
- statistical calculations
- efficient memory usage

Nearly every machine learning library internally uses NumPy.

Scikit-learn

What is it?

Scikit-learn is the standard machine learning library for Python.

It contains dozens of algorithms and utilities.

Think of it as a toolbox for machine learning.

We used it for:

Train/Test Split

Instead of training on every participant, the dataset is divided into:

- Training set
- Testing set

The model learns from one portion and is evaluated on unseen data.

This measures generalization.

Pipelines

A Pipeline combines preprocessing and model training into a single workflow.

Instead of writing

Clean

↓

Encode

↓

Scale

↓

Train

individually,

the pipeline performs everything automatically.

Advantages:

- cleaner code
 - reproducibility
 - avoids data leakage
-

SimpleImputer

Some variables had missing values.

SimpleImputer automatically replaced missing numerical values using the median.

This allowed all participants to remain usable.

OneHotEncoder

Machine learning models cannot understand words like

Male

Female

Left-handed

OneHotEncoder converts them into binary variables.

Example

Gender

Male

Female

becomes

Gender_Male

Gender_Female

StandardScaler

Some algorithms perform better when numerical variables are standardized.

It transforms variables to have:

- mean = 0
- standard deviation = 1

This prevents variables with larger numerical ranges from dominating the model.

Tree-based models like Random Forest and XGBoost do not require scaling, but algorithms like Logistic Regression and SVM often benefit from it.

Logistic Regression

What is it?

Despite its name, Logistic Regression is a classification algorithm.

It estimates the probability that a participant belongs to a class.

It assumes a linear relationship between predictors and the log-odds of the outcome.

Advantages:

- simple
- interpretable
- fast

Disadvantages:

- struggles with nonlinear relationships

It served as our baseline model.

Support Vector Machine (SVM)

What is it?

SVM tries to find the optimal boundary separating two classes.

Instead of simply drawing a line,

it searches for the boundary that maximizes the distance (margin) between the classes.

Advantages:

- excellent on many datasets
- handles high-dimensional data well

Disadvantages:

- computationally expensive
 - harder to interpret
-

Random Forest

What is it?

Random Forest is an ensemble learning algorithm.

Instead of using one decision tree,

it builds hundreds of trees.

Each tree votes.

The majority vote becomes the prediction.

Advantages:

- handles nonlinear relationships
- resistant to overfitting
- provides feature importance

This model achieved excellent performance.

XGBoost

What is it?

XGBoost stands for **Extreme Gradient Boosting**.

Instead of building independent trees like Random Forest, it builds trees sequentially.

Each new tree focuses on correcting the mistakes of previous trees.

This process gradually improves the model.

Advantages:

- state-of-the-art for structured/tabular data
- extremely accurate
- handles missing data well
- includes regularization to reduce overfitting
- computationally efficient

This became our best-performing model.

Cross Validation

Why not just split once?

One train/test split can be lucky or unlucky.

Cross-validation repeats training several times using different partitions of the data.

Example:

Fold 1

Fold 2

Fold 3

Fold 4

Fold 5

The final performance is the average across all folds.

Advantages:

- more reliable estimate
 - reduces dependence on one split
 - improves scientific validity
-

ROC-AUC

What is ROC?

ROC stands for

Receiver Operating Characteristic.

Instead of evaluating only one decision threshold,
ROC evaluates every possible threshold.

What is AUC?

Area Under the Curve.

Interpretation:

0.50

Random guessing

↓

0.70

Reasonable

↓

0.80

Good

↓

0.90+

Excellent

↓

1.00

Perfect

Our XGBoost achieved about **0.97** in cross-validation, indicating excellent discrimination between the two classes on this dataset.

Accuracy

Accuracy is simply

Correct Predictions

Total Predictions

Our final accuracy:

92.5%

However,

accuracy alone can be misleading if classes are imbalanced,

which is why ROC-AUC was emphasized.

Confusion Matrix

Instead of one number,

the confusion matrix shows

- True Positives
- True Negatives
- False Positives
- False Negatives

It tells us exactly where the model makes mistakes.

SHAP

This is probably the most important algorithm after XGBoost.

SHAP stands for

SHapley Additive exPlanations.

It originates from cooperative game theory.

Imagine several people working together to earn money.

A natural question is:

How much did each person contribute?

SHAP asks the same question for machine learning.

Instead of people,

the "players" are features:

- Age
- ADHD Index
- IQ
- Gender
- Inattentive symptoms

For every prediction,

SHAP estimates how much each feature pushed the prediction toward or away from ADHD.

Advantages:

- Explains black-box models
 - Works with tree-based algorithms like XGBoost
 - Provides both **global explanations** (important features overall) and **local explanations** (why a specific participant received a specific prediction)
-

Git & GitHub

Git

A version control system that records changes to your code, making it easy to track progress, collaborate, and revert mistakes.

GitHub

A platform for hosting Git repositories online.

For this project, GitHub serves as:

- the source code repository,
 - documentation host,
 - reproducibility record,
 - and portfolio piece that others can review and reproduce.
-

The complete workflow

ADHD-200 Dataset

↓

Pandas (load & clean data)

↓

NumPy (numerical arrays)

↓

Scikit-learn Pipeline

↓

Data preprocessing

↓

Train/Test Split

↓

Model Training

(Logistic Regression, SVM, Random Forest, XGBoost)



Cross-Validation



Performance Evaluation

(ROC-AUC, Accuracy, Confusion Matrix)



Best Model (XGBoost)



SHAP Explainability



Scientific Interpretation

This workflow demonstrates not just how to build a predictive model, but how to evaluate it rigorously and interpret its behavior in a transparent, reproducible way.

Glossary of Machine Learning and AI Terms

Algorithm

A set of mathematical instructions used by a computer to solve a problem or make predictions.

In this project, algorithms such as Random Forest and XGBoost learn patterns from clinical data to predict ADHD classification.

Artificial Intelligence (AI)

A field of computer science focused on creating systems capable of performing tasks that normally require human intelligence, such as recognizing patterns, making decisions, or learning from experience.

Machine Learning (ML)

A branch of AI where computers learn patterns from data instead of being explicitly programmed with every rule.

In this project, the model learns relationships between participant characteristics and ADHD classification.

Feature

An input variable used by a machine learning model to make predictions.

Examples from this project:

- Age
- Gender
- IQ measurements
- ADHD Index
- Inattentive symptoms

Features are the information the model uses to learn.

Target Variable (Label)

The outcome that the model attempts to predict.

In this project:

- 0 = No ADHD
- 1 = ADHD

The target variable is also called the label or outcome variable.

Dataset

A structured collection of information used for analysis and machine learning.

The ADHD-200 dataset contains participant information, including demographic, cognitive, and symptom-related measurements.

Sample / Participant

One individual observation in a dataset.

In this project, each row represents one participant from the ADHD-200 dataset.

Classification

A machine learning task where the goal is to assign data into categories.

Example:

Participant → ADHD

Participant → No ADHD

This project is a binary classification problem because there are two possible outcomes.

Binary Classification

A classification problem with exactly two possible classes.

Example:

Class 0: No ADHD

Training Data

The portion of the dataset used to teach the model.

The model analyzes training examples and learns relationships between features and outcomes.

Testing Data

A separate portion of the dataset that the model has never seen during training.

It is used to evaluate whether the model learned general patterns rather than memorized the training data.

Model

A mathematical system that learns relationships from data and uses them to make predictions.

Examples:

- Logistic Regression
 - Random Forest
 - XGBoost
-

Training

The process where a machine learning algorithm learns patterns from examples.

During training, the model adjusts its internal parameters to improve prediction accuracy.

Prediction

The output generated by a trained machine learning model.

Example:

A participant's data is given to the model, and it predicts:

Probability of ADHD: 82%

Preprocessing

The preparation of raw data before it can be used by machine learning algorithms.

Examples:

- Cleaning missing values
 - Encoding categories
 - Scaling numerical variables
-

Missing Values

Data points that are unavailable or not recorded.

Examples:

- Missing IQ measurement
- Missing symptom score

Machine learning algorithms usually require strategies to handle missing information.

Imputation

A method for replacing missing values with estimated values.

Example:

Replacing missing numerical values with the median value of that feature.

Encoding

The process of converting non-numerical information into numbers that machine learning algorithms can understand.

Example:

Before:

Gender = Male

After encoding:

Gender_Male = 1

Gender_Female = 0

Feature Scaling

A process that puts numerical variables into a comparable range.

For example:

Age and IQ have different numerical scales.

Scaling prevents some algorithms from giving too much importance to variables simply because their numbers are larger.

Pipeline

A sequence of automated processing steps connected together.

Example:

Raw Data

↓

Cleaning

↓

Encoding

↓

Scaling

↓

Machine Learning Model

Pipelines improve reproducibility and reduce mistakes.

Overfitting

A situation where a model learns the training data too specifically, including noise or random patterns.

An overfitted model performs well on training data but poorly on new data.

Generalization

The ability of a model to perform well on new, unseen data.

A successful machine learning model learns general patterns rather than memorizing examples.

Ensemble Learning

A technique where multiple models are combined to create a stronger model.

Instead of relying on one decision-maker, the system combines many.

Examples:

- Random Forest
 - XGBoost
-

Decision Tree

A model that makes predictions by asking a sequence of questions.

Example:

Is ADHD Index high?

↓

Does symptom score exceed threshold?

↓

Prediction

Random Forest

An ensemble algorithm that combines many decision trees.

Each tree makes a prediction, and the final decision is based on the combined results.

Gradient Boosting

A machine learning technique where models are built sequentially, with each new model attempting to correct previous errors.

XGBoost

Extreme Gradient Boosting.

A highly optimized gradient boosting algorithm commonly used for structured datasets.

It was the best-performing model in this project.

Logistic Regression

A statistical classification algorithm that estimates the probability of belonging to a class.

It was used as a simple baseline model.

Support Vector Machine (SVM)

A machine learning algorithm that finds the best boundary separating different classes.

It is effective when complex relationships exist between variables.

Cross-Validation

A method for testing model reliability by repeatedly training and evaluating the model using different parts of the dataset.

It provides a more reliable estimate of performance.

Metric

A measurement used to evaluate model performance.

Examples:

- Accuracy
 - ROC-AUC
 - Precision
 - Recall
-

Accuracy

The percentage of predictions that are correct.

Formula:

Correct Predictions / Total Predictions

Precision

Measures how many positive predictions were actually correct.

Example:

Of all participants predicted to have ADHD, how many truly had ADHD?

Recall

Measures how many actual positive cases were successfully detected.

Example:

Of all participants with ADHD, how many did the model identify?

F1 Score

A balance between precision and recall.

It is useful when classes are not perfectly balanced.

Confusion Matrix

A table showing correct and incorrect classifications.

It separates:

- True positives
 - True negatives
 - False positives
 - False negatives
-

ROC Curve

A graph showing model performance across different classification thresholds.

ROC-AUC

Area Under the Receiver Operating Characteristic Curve.

It measures how well a model separates two classes.

General interpretation:

- 0.5 = random guessing

- 1.0 = perfect separation
-

Explainable AI (XAI)

Methods that help humans understand why AI models make specific predictions.

This is especially important in healthcare applications.

SHAP

SHapley Additive exPlanations.

A method that estimates how much each feature contributes to a model prediction.

It answers:

"Why did the model make this decision?"

Feature Importance

A measurement of how strongly each variable influences model predictions.

Model Interpretability

The ability to understand and explain how a model reaches its decisions.

Reproducibility

The ability for another person to repeat the analysis and obtain similar results using the same data and code.

This is a fundamental principle of scientific research.

Closing Remark

Artificial intelligence is not simply about creating models that achieve high accuracy. A successful AI project requires understanding the entire journey: from collecting and preparing data, to selecting appropriate algorithms, evaluating performance, and interpreting the results responsibly.

This project demonstrates a complete machine learning workflow applied to a healthcare-related research question. By combining predictive modeling with explainable AI, it moves beyond the question of **"Can the model predict?"** and addresses the equally important question of **"Why does the model predict?"**

For beginners, the most important lesson is that AI is not a mysterious black box. It is a collection of mathematical tools, engineering practices, and scientific methods that allow us to extract meaningful patterns from complex information.

The future of healthcare AI will not depend only on more powerful algorithms, but on developing systems that are **accurate, transparent, reproducible, and scientifically responsible**.